

Exploratory Analysis of DePaul University — International Student Applications

13 April, 2026 | Impact Makers Project

01 Dataset Familiarization Summary

Initial Observations of Dataset

Dataset resembles a university admissions CRM export Contains:
Student demographics (name, country, city).
Academic background (college, intake) Application tracking data large number of columns are Completely empty and Redundant or duplicated. Data is not analysis-ready and requires cleaning.

Dataset Structure (Simple View)

Metric	Value / Finding
Total Records	7,543 student application rows
Total Columns	80 variables across the dataset
Numerical Variables	50 columns (timestamps, IDs, counts)
Text Variables	30 columns (names, countries, comments)
Completely Empty Columns	~30+ columns with no usable data
Duplicate / Redundant Columns	Multiple fields repeated 2–3 times (e.g., University, Modified_At, Created_At)

02 Executive Summary

This project analyses the International Student Application Records of DePaul University dataset), which contains 7,543 records and 80 variables describing international student applications.

The primary objective of this analysis is to clean, document, and explore the dataset in order to better understand the structure and patterns within international student applications. Particular attention is given to data quality issues, including duplicate applicant records, missing values, and variables originating from multiple integrated systems.

To support the analysis, Python is used for data cleaning, transformation, and exploratory analysis. A data dictionary is also developed to document each variable and improve the interpretability of the dataset.

The outcome of this project is a cleaner, well- documented dataset and an initial set of insights that can support further analysis of international student recruitment and application trends.

03 Dataset Description

The dataset contains international student application records from DePaul University. It has 7,543 rows and 80 variables. The data includes information about the applicants, such as Given Name, Last-named, Date_of_Birth, Phone Number, and Country. It also includes academic program preferences like College and Major_1st_Choice, as well as admission information such as Admit Date, Is Admitted, and Is_Global_Grad. In addition, the dataset contains communication history and administrative system information related to the application process.

Data Dictionary

A well-structured data dictionary describing the dataset was created and includes the following information:

- S/N
- Field Name
- Data Types
- Variable Definition
- Note

[Here's a link to this well-documented data dictionary:](https://docs.google.com/spreadsheets/d/1aFQsgo)

https://docs.google.com/spreadsheets/d/1aFQsgomwdVA26im0JYV5dKtCgVOEMRdCop2gfFXdu6U/e_dit?usp=sharing

03 Data Quality Report

After a thorough initial exploration of the dataset, the following issues have been discovered:

- There are multiple duplicates of Reference_ID which is meant to be the unique identifier for each applicant.
- There are missing or NULL values across several variables.
- Some columns contain only NULL values. Examples:
- Major, Degree_Type, Citizenship → completely empty
- Street_2 → ~60% missing
- Region → ~12% missing
- There are inconsistencies in the data format of some columns and in categorical values of other columns.
- Some columns have been merged together and would require splitting.

Anomalies

The dataset contains several structural and quality issues that affect its usability. A large number of columns are completely empty, making them irrelevant

for analysis. There are duplicate records and repeated columns (e.g., with “-2” and “-3” suffixes), which introduce redundancy and potential inaccuracies.

Data inconsistency is also observed in text fields such as country and program names due to variations in formatting. Some columns contain multiple values in a single field, violating proper data structure.

Additionally, timestamps are stored as integers instead of standard date formats, limiting time-based analysis.

The dataset mixes student information with CRM-related interaction data, leading to inconsistent data granularity. Unstructured text fields and incomplete contact information further reduce data reliability.

Overall, these anomalies highlight the need for thorough data cleaning and standardization before meaningful analysis can be performed.

04 Exploratory Data Analysis

Numerical Variables

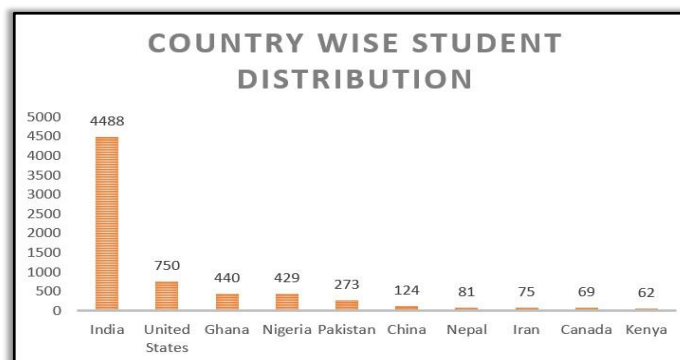
Most numeric variables are IDs or timestamps → limited analytical value.

Example: Reference_ID

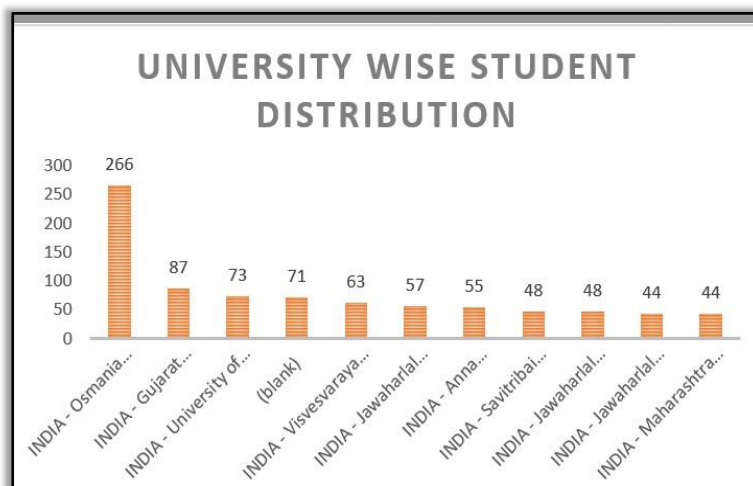
- Mean: ~473M
- Median: ~469M
- Min: ~291K
- Max: ~999M

Pure identifier not useful insights

Visualization

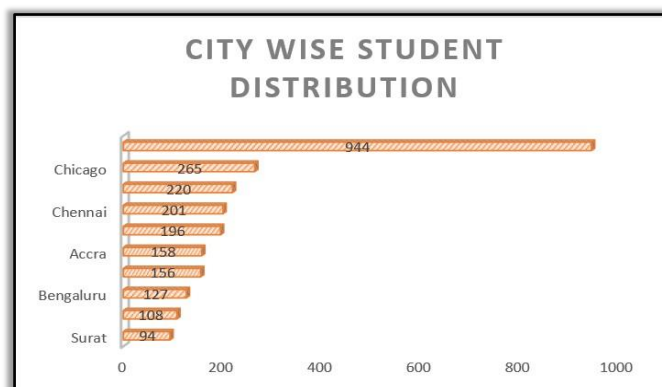


1. India dominates overwhelmingly with 4,488 students, contributing the vast majority of the population.
 2. The United States (750) is a distant second, followed by Ghana (440) and Nigeria (429).
 3. Other countries like Pakistan (273), China (124), and Nepal (81) contribute relatively small numbers.
 4. Countries such as Iran, Canada, and Kenya have minimal representation (below 100 each)
- Insight: The student base is heavily concentrated in India, indicating strong recruitment or outreach there.



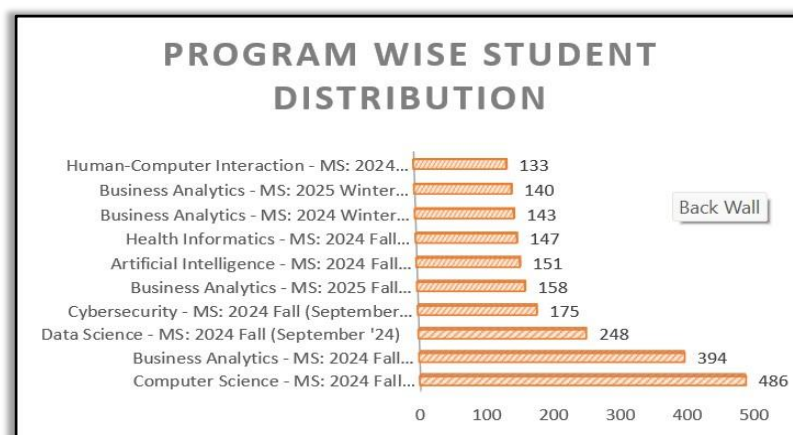
1. One institution (INDIA – Osmania University) leads with 266 students, significantly ahead of others.
2. The next group of universities contributes between 40–90 students each, showing a steep drop after the top institution.
3. A long tail of universities contributes smaller but consistent numbers.

Insight: Student intake is somewhat concentrated in a few key universities, but overall distributed across many institutions.



1. One city stands out with 944 students, far exceeding all others.
2. Secondary cities like Chicago (265) and Chennai (220) show moderate representation.
3. Cities such as Bengaluru, Accra, and Surat have smaller but notable contributions (below 200).

Insight: There is a strong geographic concentration in one primary city, with a gradual drop across others.



1. Computer Science (486 students) is the most popular program.
2. Followed by Business Analytics (394) and Data Science (248).
3. Mid-tier programs include Cybersecurity (175) and AI (151).
4. Programs like Health Informatics and HCI have the lowest enrollments (~130–150 range).

Insight: Tech-focused programs dominate, especially Computer Science and Analytics fields. Specialized or niche programs attract fewer students.

05 Overall Takeaways & Key Insights

- Geographic Concentration: Strong skew toward India and a few key cities.
- Program Preference: Clear demand for technology and data-driven programs.
- Diversity Opportunity: There is potential to increase representation from other countries and cities.
- Recruitment Strategy: Likely reliant on specific universities and regions, which could be expanded for balance.